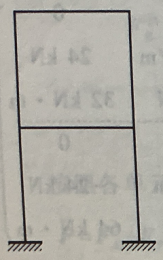
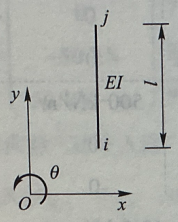
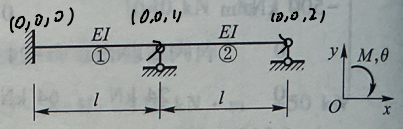
11-3 试求图示连续梁的结构刚度矩阵 K，已知杆件 EI 为常数。



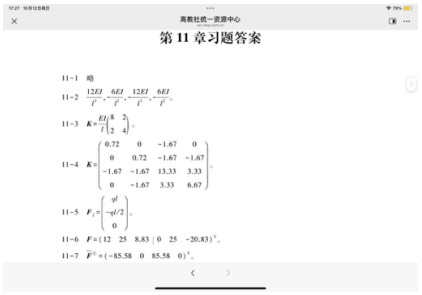
题 11-1 图



题 11-2 图



题 11-3 图



第 11 章习题答案

11-1 略

11-2 $\begin{pmatrix} \frac{12EI}{l^3} & -\frac{6EI}{l^2} \\ -\frac{12EI}{l^3} & -\frac{6EI}{l^2} \end{pmatrix}$。

11-3 $K=\frac{EI}{l}\begin{pmatrix} 8 & 2 \\ 2 & 4 \end{pmatrix}$。

11-4 $K=\begin{pmatrix} 0.72 & 0 & -1.67 & 0 \\ 0 & 0.72 & -1.67 & -1.67 \\ -1.67 & -1.67 & 13.33 & 3.33 \\ 0 & -1.67 & 3.33 & 6.67 \end{pmatrix}$。

11-5 $F_2=\begin{pmatrix} ql \\ -q/2 \\ 0 \end{pmatrix}$。

11-6 $F=(12 \quad 25 \quad 8.83 \mid 0 \quad 25 \quad -20.83)^T$。

11-7 $\bar{F}^{\oplus}=(-85.58 \quad 0 \quad 85.58 \quad 0)^T$。

解：

① $k^{①}=\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ & & & & & 0 \\ & & & & & 0 \\ & & & & & 0 \\ & & & & & 0 \\ & & & & & 1 \end{bmatrix}$ 行 $\begin{matrix} & & & & & k_{66} \end{matrix}$

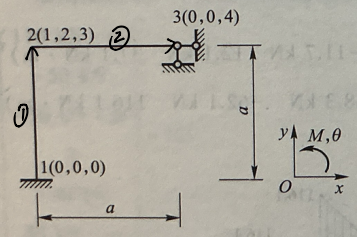
$k=\begin{bmatrix} k_{66}^{①}+k_{33}^{①} & k_{36}^{②} \\ k_{63}^{②} & k_{66}^{②} \end{bmatrix}=\begin{bmatrix} \frac{3EI}{l} & \frac{2EI}{l} \\ \frac{2EI}{l} & \frac{4EI}{l} \end{bmatrix}$

$k^{②}=\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 2 \\ & & & & & 0 \\ & & & & & 0 \\ & & k_{33} & & k_{36} & 1 \\ & & & & & 0 \\ & & & & & 0 \\ & & k_{63} & & k_{66} & 2 \end{bmatrix}$

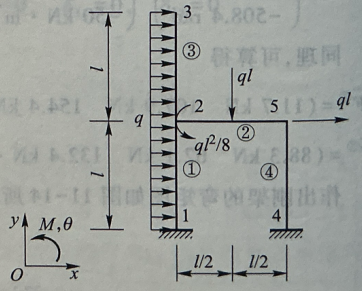
11-4

11-4　试求图示刚架的结构刚度矩阵 $\boldsymbol{K}$。已知 $EI=1\times10^{6}\ \mathrm{kN\cdot m}$，$EA=1\times10^{5}\ \mathrm{kN}$，$a=6\ \mathrm{m}$。

11-5　试求图示结构结点 2 的综合荷载列阵 $\boldsymbol{F}_2$。



题 11-4 图



题 11-5 图

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第 11 章习题答案

11-1　略

11-2　$\frac{12EI}{l^3}, -\frac{6EI}{l^2}, -\frac{12EI}{l^3}, -\frac{6EI}{l^2}$。

11-3　$\boldsymbol{K}=\frac{EI}{l}\begin{pmatrix}8 & 2\\ 2 & 4\end{pmatrix}$。

11-4　$\boldsymbol{K}=\begin{pmatrix}0.72 & 0 & -1.67 & 0\\ 0 & 0.72 & -1.67 & -1.67\\ -1.67 & -1.67 & 13.33 & 3.33\\ 0 & -1.67 & 3.33 & 6.67\end{pmatrix}$。

11-5　$\boldsymbol{F}_2=\begin{pmatrix}ql\\ -ql/2\\ 0\end{pmatrix}$。

11-6　$\boldsymbol{F}=(12\quad 25\quad 8.83\ \vdots\ 0\quad 25\quad -20.83)^{\mathrm{T}}$。

11-7　$\overline{\boldsymbol{F}}^{①}=(-85.58\quad 0\quad 85.58\quad 0)^{\mathrm{T}}$。

解：$\bar{k}^{①}=$ 0 0 0 1 2 3

$\bar{k}^{②}=$ 1 2 3 0 0 4 $=k^{②}$

$k=\begin{bmatrix} k_{44}^{①}+k_{11}^{②} & k_{45}^{①}+k_{12}^{②} & k_{46}^{①}+k_{13}^{②} & k_{14}^{②}\\ k_{54}^{①}+k_{21}^{②} & k_{55}^{①}+k_{22}^{②} & k_{56}^{①}+k_{23}^{②} & k_{24}^{②}\\ k_{64}^{①}+k_{31}^{②} & k_{65}^{①}+k_{32}^{②} & k_{66}^{①}+k_{33}^{②} & k_{34}^{②}\\ k_{41}^{②} & k_{42}^{②} & k_{43}^{②} & k_{44}^{②}\end{bmatrix}$

$\theta=90^{\circ},\ \cos\theta=0,\ \sin\theta=1$

$k^{①}=T^{\mathrm{T}}\bar{k}T$

$T=\begin{bmatrix}\cos\theta & \sin\theta & 0 & 0 & 0 & 0\\ -\sin\theta & \cos\theta & 0 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 0 & \cos\theta & \sin\theta & 0\\ 0 & 0 & 0 & -\sin\theta & \cos\theta & 0\\ 0 & 0 & 0 & 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}0 & 1 & 0 & 0 & 0 & 0\\ -1 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & -1 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 1\end{bmatrix}=T^{\mathrm{T}}$

$k^{①}=\begin{bmatrix}0 & -1 & 0 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & -1 & 0\\ 0 & 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}k_{44} & k_{45} & k_{46} & 0 & 0 & 0\\ k_{54} & k_{55} & k_{56} & 0 & 0 & 0\\ k_{64} & k_{65} & k_{66} & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0\end{bmatrix}\begin{bmatrix}0 & 1 & 0 & 0 & 0 & 0\\ -1 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & -1 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 1\end{bmatrix}$

$=\begin{bmatrix}0 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & -k_{54} & -k_{55} & -k_{56}\\ 0 & 0 & 0 & +k_{44} & +k_{45} & +k_{46}\\ 0 & 0 & 0 & k_{64} & k_{65} & k_{66}\end{bmatrix}\begin{bmatrix}0 & 1 & 0 & 0 & 0 & 0\\ -1 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & -1 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 1\end{bmatrix}$

$$= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & k_{55} & -k_{54} & -k_{56} \\ 0 & 0 & 0 & -k_{45} & k_{44} & k_{46} \\ 0 & 0 & 0 & -k_{65} & k_{64} & k_{66} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ & \\ 0 & \end{bmatrix}$$

$$\begin{bmatrix} \frac{12EI}{l^3} & 0 & -\frac{6EI}{l^2} \\ 0 & \frac{EA}{l} & 0 \\ -\frac{6EI}{l^2} & 0 & \frac{4EI}{l} \end{bmatrix}$$

$$K = \begin{bmatrix} k_{44}^{(1)} + k_{11}^{(2)} & k_{45}^{(1)} + k_{12}^{(2)} & k_{46}^{(1)} + k_{13}^{(2)} & k_{16}^{(2)} \\ k_{54}^{(1)} + k_{21}^{(2)} & k_{55}^{(1)} + k_{22}^{(2)} & k_{56}^{(1)} + k_{23}^{(2)} & k_{26}^{(2)} \\ k_{64}^{(1)} + k_{31}^{(2)} & k_{65}^{(1)} + k_{32}^{(2)} & k_{66}^{(1)} + k_{33}^{(2)} & k_{36}^{(2)} \\ k_{61}^{(2)} & k_{62}^{(2)} & k_{63}^{(2)} & k_{66}^{(2)} \end{bmatrix} = \begin{bmatrix} \frac{12EI}{l^3} + \frac{EA}{l} & 0+0 & -\frac{6EI}{l^2} + 0 & 0 \\ 0+0 & \frac{EA}{l} + \frac{12EI}{l^3} & -\frac{6EI}{l^2} + \frac{6EI}{l^2} & \frac{6EI}{l^2} \\ -\frac{6EI}{l^2} + 0 & 0 + \frac{6EI}{l^2} & \frac{4EI}{l} + \frac{4EI}{l} & \frac{2EI}{l} \\ 0 & \frac{6EI}{l^2} & \frac{2EI}{l} & \frac{4EI}{l} \end{bmatrix}$$

$$= \begin{bmatrix} 0.72 & 0 & -1.67 & 0 \\ 0 & 0.72 & 1.67 & 1.67 \\ -1.67 & 1.67 & 13.33 & 3.33 \\ 0 & 1.67 & 3.33 & 6.67 \end{bmatrix}$$

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第 11 章习题答案

11-1 略

11-2 $\frac{12EI}{l^3}, -\frac{6EI}{l^2}, -\frac{12EI}{l^3}, -\frac{6EI}{l^2}$。

11-3 $\boldsymbol{K} = \frac{EI}{l}\begin{pmatrix} 8 & 2 \\ 2 & 4 \end{pmatrix}$。

11-4 $\boldsymbol{K} = \begin{pmatrix} 0.72 & 0 & -1.67 & 0 \\ 0 & 0.72 & -1.67 & -1.67 \\ -1.67 & -1.67 & 13.33 & 3.33 \\ 0 & -1.67 & 3.33 & 6.67 \end{pmatrix}$。

11-5 $\boldsymbol{F}_2 = \begin{pmatrix} ql \\ -ql/2 \\ 0 \end{pmatrix}$。

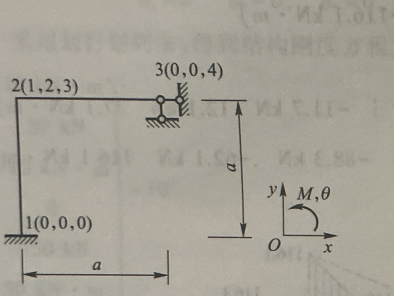
11-6 $\boldsymbol{F} = (12 \quad 25 \quad 8.83 \vdots 0 \quad 25 \quad -20.83)^{\mathrm{T}}$。

11-7 $\overline{\boldsymbol{F}}^{(1)} = (-85.58 \quad 0 \quad 85.58 \quad 0)^{\mathrm{T}}$。

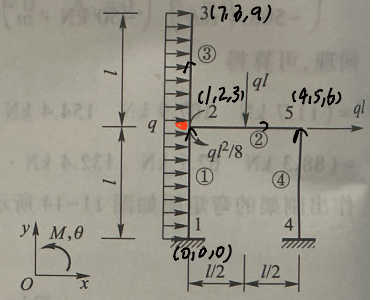
2025年10月14日

11-5

11-5 试求图示结构结点2的综合荷载列阵 F_2。



题 11-4 图



题 11-5 图

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第 11 章习题答案

11-1 略

11-2 $\frac{12EI}{l^3}, -\frac{6EI}{l^2}, -\frac{12EI}{l^3}, -\frac{6EI}{l^2}$。

11-3 $\boldsymbol{K}=\frac{EI}{l}\begin{pmatrix}8 & 2\\ 2 & 4\end{pmatrix}$。

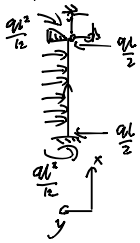
11-4 $\boldsymbol{K}=\begin{pmatrix}0.72 & 0 & -1.67 & 0\\ 0 & 0.72 & -1.67 & -1.67\\ -1.67 & -1.67 & 13.33 & 3.33\\ 0 & -1.67 & 3.33 & 6.67\end{pmatrix}$。

11-5 $\boldsymbol{F}_2=\begin{pmatrix}ql\\ -ql/2\\ 0\end{pmatrix}$。

11-6 $\boldsymbol{F}=(12 \quad 25 \quad 8.83 \mid 0 \quad 25 \quad -20.83)^{\mathrm{T}}$。

11-7 $\overline{\boldsymbol{F}}^{\mathrm{b}}=(-85.58 \quad 0 \quad 85.58 \quad 0)^{\mathrm{T}}$。

解：对结构①

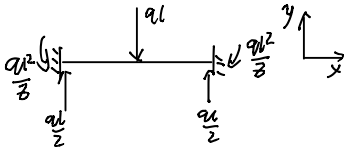


$\overline{F}_F^{①}=(0 \quad \frac{ql}{2} \quad \frac{ql^2}{12} \mid 0 \quad \frac{ql}{2} \quad -\frac{ql^2}{12})^T$ 则 $T^{①T}=\begin{bmatrix}0&-1&0&\vdots&[0]\\1&0&0&&\\0&0&1&&\\\cdots&\cdots&\cdots&\cdots&\cdots\\[0]&&&\vdots&\begin{matrix}0&-1&0\\1&0&0\\0&0&1\end{matrix}\end{bmatrix}$

$\theta^{①}=90°,\ \cos\theta=0,\ \sin\theta=1$

$$F_E^{①}=-T^{①T}\overline{F}_F^{①}=-\begin{bmatrix}0&-1&0&\vdots&[0]\\1&0&0&&\\0&0&1&&\\\cdots&\cdots&\cdots&\cdots&\cdots\\[0]&&&\vdots&\begin{matrix}0&-1&0\\1&0&0\\0&0&1\end{matrix}\end{bmatrix}\begin{bmatrix}0\\\frac{ql}{2}\\\frac{ql^2}{12}\\0\\\frac{ql}{2}\\-\frac{ql^2}{12}\end{bmatrix}=\begin{bmatrix}\frac{ql}{2}&0&-\frac{ql^2}{12}&;&\frac{ql}{2}&0&\frac{ql^2}{12}\end{bmatrix}^T$$

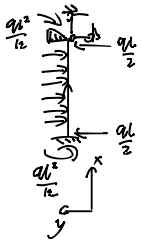
对结构②



$\overline{F}_F^{②}=[0 \quad \frac{ql}{2} \quad \frac{ql^2}{8} ; 0 \quad \frac{ql}{2} \quad -\frac{ql^2}{8}]$

$F_E^{②}=-T^{②T}\overline{F}_F^{②}$ 且 $\theta=0,\ \sin\theta=0,\ \cos\theta=1$ 则 $\overline{F}_E^{②}=[0 \quad -\frac{ql}{2} \quad -\frac{ql^2}{8} ; 0 \quad \frac{-ql}{2} \quad \frac{ql^2}{8}]^T$

对结构③



$\overline{F}_F^{③}=(0 \quad \frac{ql}{2} \quad \frac{ql^2}{12} \mid 0 \quad \frac{ql}{2} \quad -\frac{ql^2}{12})^T$ 则 $T^{③T}=\begin{bmatrix}0&-1&0&\vdots&[0]\\1&0&0&&\\0&0&1&&\\\cdots&\cdots&\cdots&\cdots&\cdots\\[0]&&&\vdots&\begin{matrix}0&-1&0\\1&0&0\\0&0&1\end{matrix}\end{bmatrix}$

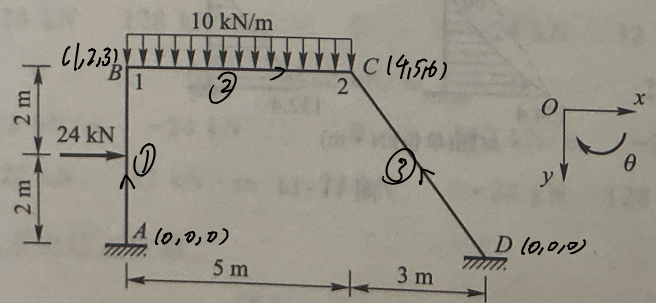
$\theta^{③}=90°,\ \cos\theta=0,\ \sin\theta=1$

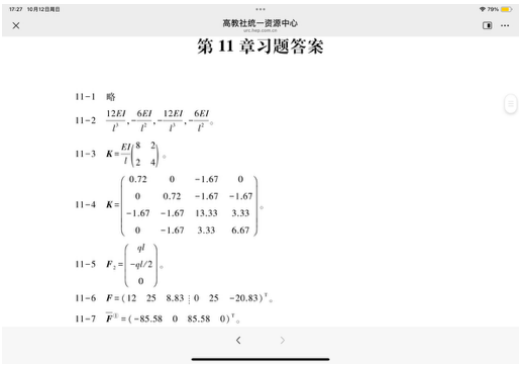
$$F_E^{③}=-T^{③T}\overline{F}_F^{③}=-\begin{bmatrix}0&-1&0&\vdots&[0]\\1&0&0&&\\0&0&1&&\\\cdots&\cdots&\cdots&\cdots&\cdots\\[0]&&&\vdots&\begin{matrix}0&-1&0\\1&0&0\\0&0&1\end{matrix}\end{bmatrix}\begin{bmatrix}0\\\frac{ql}{2}\\\frac{ql^2}{12}\\0\\\frac{ql}{2}\\-\frac{ql^2}{12}\end{bmatrix}=\begin{bmatrix}\frac{ql}{2}&0&-\frac{ql^2}{12}&;&\frac{ql}{2}&0&\frac{ql^2}{12}\end{bmatrix}^T$$

$$F_{E2}=\begin{bmatrix}\frac{ql}{2}\\0\\\frac{ql^2}{12}\end{bmatrix}+\begin{bmatrix}0\\-\frac{ql}{2}\\-\frac{ql^2}{8}\end{bmatrix}+\begin{bmatrix}\frac{ql}{2}\\0\\\frac{-ql^2}{12}\end{bmatrix}=\begin{bmatrix}ql&-\frac{ql}{2}&-\frac{ql^2}{8}\end{bmatrix}^T$$

$$F_2=F_{E2}+F_P=\begin{bmatrix}ql\\\frac{ql}{2}\\-\frac{ql^2}{8}\end{bmatrix}+\begin{bmatrix}0\\0\\\frac{ql^2}{8}\end{bmatrix}=\begin{bmatrix}ql\\-\frac{ql}{2}\\0\end{bmatrix}$$

11-6　试求图示结构的综合荷载列阵。





第 11 章习题答案

11-1　略

11-2　$\frac{12EI}{l^3}, \frac{6EI}{l^2}, \frac{12EI}{l^3}, \frac{6EI}{l^2}$。

11-3　$K=\frac{EI}{l}\begin{pmatrix}8 & 2\\ 2 & 4\end{pmatrix}$。

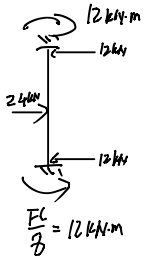
11-4　$K=\begin{pmatrix}0.72 & 0 & -1.67 & 0\\ 0 & 0.72 & -1.67 & -1.67\\ -1.67 & -1.67 & 13.33 & 3.33\\ 0 & -1.67 & 3.33 & 6.67\end{pmatrix}$。

11-5　$F_1=\begin{pmatrix}ql\\ -ql/2\\ 0\end{pmatrix}$。

11-6　$F=(12\quad 25\quad 8.83\mid 0\quad 25\quad -20.83)^{\mathrm{T}}$。

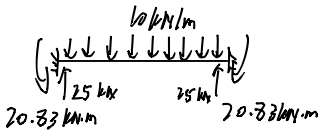
11-7　$\bar{F}^{①}=(-85.58\quad 0\quad 85.58\quad 0)^{\mathrm{T}}$。

解：结构①



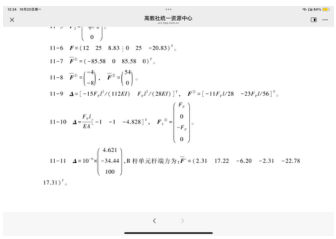
$$\bar{F}_F^{①}=\begin{bmatrix}-12\text{kN}\\ 0\\ -12\text{kN·m}\\ -12\text{kN}\\ 0\\ 12\text{kN·m}\end{bmatrix}\begin{matrix}0\\0\\0\\1\\2\\3\end{matrix}$$

结构②：



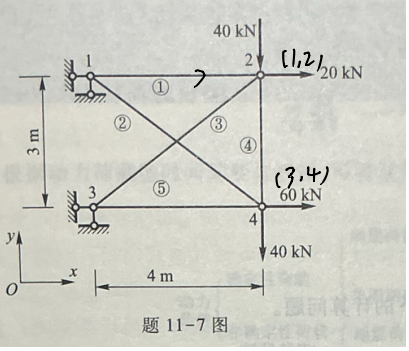
$$\bar{F}_{F②}=\begin{bmatrix}0\\-25\text{ kN}\\-20.83\text{kN·m}\\0\\-25\text{kN}\\20.83\text{kN·m}\end{bmatrix}\begin{matrix}1\\2\\3\\4\\5\\6\end{matrix}$$

$$\bar{F}_E=-\bar{\Sigma}\bar{F}_F=\begin{bmatrix}12\\25\\8.33\\0\\25\\-20.83\end{bmatrix}\qquad \bar{F}=\bar{F}_E+\bar{F}_P=\begin{bmatrix}12\\25\\8.33\\0\\25\\-20.83\end{bmatrix}$$



2025年10月22日

11-7　11-7　图示桁架的结点位移列阵 $\boldsymbol{\Delta}=\frac{1}{EA}(342.322\quad -1\ 139.555\quad -137.680\quad 1\ 167.111)^{\mathrm{T}}$，已知各杆 EA 为常数，试求单元①的内力。



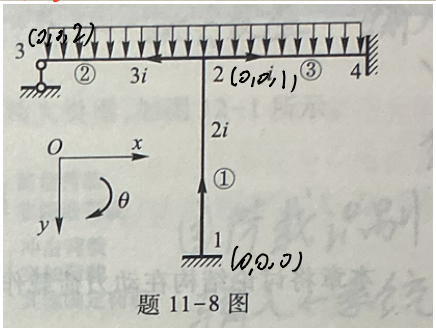
题 11-7 图

解：$\bar{F}^{①}=k^{①}\delta^{①}$

$$\bar{F}^{①}=\begin{bmatrix}\bar{F}_{1x}^{①}\\ \bar{F}_{1y}^{①}\\ \bar{F}_{2x}^{①}\\ \bar{F}_{2y}^{①}\end{bmatrix}=\frac{EA}{l}\begin{bmatrix}1 & 0 & -1 & 0\\ 0 & 0 & 0 & 0\\ -1 & 0 & 1 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}\frac{1}{EA}\begin{bmatrix}0\\0\\342.322\\-1139.555\end{bmatrix}=\begin{bmatrix}-35.58\\0\\35.58\\0\end{bmatrix}$$

11-8　已知图示刚架结点的转角为 $\theta_2=-\frac{1}{i}$（逆时针），$\theta_3=\frac{4}{i}$（顺时针），各杆长度 $l=6$ m，$q=14$ kN/m，试求单元①和②的杆端力。

11-8

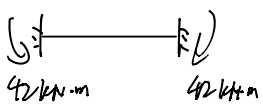


题 11-8 图

解：忽略轴向位移

对结构①：$\bar{F}_①=\bar{F}^{①}\bar{\delta}^{①}=\frac{EI}{l}\begin{bmatrix}-8 & 4\\ 4 & 8\end{bmatrix}\begin{bmatrix}0\\ -\frac{l}{EI}\end{bmatrix}=\begin{bmatrix}-4\\ -8\end{bmatrix}$

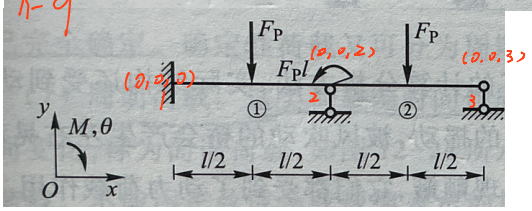
对结构②：$\bar{F}_②=\bar{k}^{②}\bar{\delta}^{②}+\bar{F}_F^{②}=\frac{EI}{l}\begin{bmatrix}12 & 6\\ 6 & 12\end{bmatrix}\begin{bmatrix}-\frac{l}{EI}\\ \frac{4l}{EI}\end{bmatrix}+\begin{bmatrix}42\\ -42\end{bmatrix}=\begin{bmatrix}54\\ 0\end{bmatrix}$



2025年11月1日

11-9 试用矩阵位移法计算图示连续梁的内力。已知 EI 为常数。

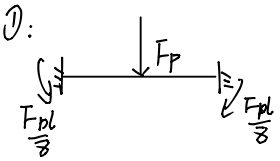
11-9



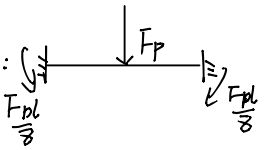
解：$i=\frac{EI}{l}$　$k^{①}=k^{②}=\begin{bmatrix}4i & 2i\\ 2i & 4i\end{bmatrix}$　对①单元 $k^{①}=\begin{bmatrix}0 & 2\\ 4i & 2i\\ 2i & 4i\end{bmatrix}\begin{matrix}0\\ 2\end{matrix}$　对②单元 $k^{②}=\begin{bmatrix}2 & 3\\ 4i & 2i\\ 2i & 4i\end{bmatrix}\begin{matrix}2\\ 3\end{matrix}$　$k=\begin{bmatrix}2 & 3\\ 8i & 2i\\ 2i & 4i\end{bmatrix}\begin{matrix}2\\ 3\end{matrix}$

$F=\Delta\cdot k$ 即 $\begin{bmatrix}M_2\\ M_3\end{bmatrix}=\begin{bmatrix}8i & 2i\\ 2i & 4i\end{bmatrix}\begin{bmatrix}\theta_1\\ \theta_2\end{bmatrix}$

对单元①：



对单元②：



即 $F=\sum F_E+\sum F_P=\begin{bmatrix}-F_P l\\ \\ -\frac{F_P l}{8}\end{bmatrix}$

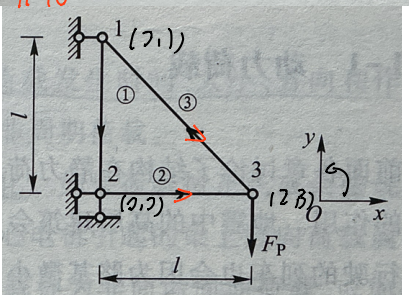
故 $\begin{bmatrix}-F_P l\\ \\ -\frac{F_P l}{8}\end{bmatrix}=\frac{EI}{l}\begin{bmatrix}8 & 2\\ \\ 2 & 4\end{bmatrix}\begin{bmatrix}\theta_2\\ \\ \theta_3\end{bmatrix}$

即 $\begin{cases}\theta_2=-\frac{15}{112}\frac{F_P l^2}{EI}\\ \theta_3=\frac{F_P l^2}{28EI}\end{cases}$

$F^{①}=k^{①}\delta^{①}+F_F^{①}=\begin{bmatrix}4 & 2\\ 2 & 4\end{bmatrix}\frac{EI}{l}\begin{bmatrix}0\\ -\frac{15}{112}\frac{F_P l^2}{EI}\end{bmatrix}+\begin{bmatrix}-\frac{F_P l}{8}\\ \\ \frac{F_P l}{8}\end{bmatrix}=\begin{bmatrix}-\frac{11}{28}F_P l\\ \\ -\frac{23F_P l}{56}\end{bmatrix}$

$F^{②}=k^{②}\delta^{②}+F_F^{②}=\begin{bmatrix}4 & 2\\ \\ 2 & 4\end{bmatrix}\frac{EI}{l}\begin{bmatrix}-\frac{15}{112}\frac{F_P l^2}{EI}\\ \\ \frac{F_P l^2}{28EI}\end{bmatrix}+\begin{bmatrix}-\frac{F_P l}{8}\\ \\ \frac{F_P l}{8}\end{bmatrix}=\begin{bmatrix}-\frac{33}{56}F_P l\\ \\ 0\end{bmatrix}$

11-10 试用矩阵位移法计算图示桁架的内力。已知 EA 为常数。



桁架 F_P 均为0,

解: 由图可知 $\theta_1=-90°$, $\theta_2=-45°$

$$\overline{k}^{①}=k^{①}=\frac{EA}{l}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{matrix}0\\0\\2\\3\end{matrix}$$

由 $\theta_1=-90°$ 则 $\cos\theta_1=0$ $\sin\theta_1=-1$

$$k^{①}=\frac{EA}{l}\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}\begin{matrix}0\\1\\0\\0\end{matrix}$$

由 $\theta_2=-45°$ 则 $\cos\theta_2=\frac{\sqrt{2}}{2}$ $\sin\theta_2=-\frac{\sqrt{2}}{2}$

$$k^{③}=\frac{EA}{\sqrt{2}l}\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}\begin{matrix}0\\1\\2\\3\end{matrix}$$

$$k=k^{①}+k^{②}+k^{③}=\frac{EA}{l}\begin{bmatrix} 1+\frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & -\frac{1}{2\sqrt{2}} \\ \frac{1}{2\sqrt{2}} & 1+\frac{1}{2\sqrt{2}} & -\frac{1}{2\sqrt{2}} \\ -\frac{1}{2\sqrt{2}} & -\frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} \end{bmatrix}\begin{matrix}1\\2\\3\end{matrix}$$

由 $F=k\Delta$

$$\begin{bmatrix} 0 \\ 0 \\ -F_P \end{bmatrix}=\frac{EA}{l}\begin{bmatrix} 1+\frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & -\frac{1}{2\sqrt{2}} \\ \frac{1}{2\sqrt{2}} & 1+\frac{1}{2\sqrt{2}} & -\frac{1}{2\sqrt{2}} \\ -\frac{1}{2\sqrt{2}} & -\frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} \end{bmatrix}\begin{bmatrix} \Delta_1 \\ \Delta_2 \\ \Delta_3 \end{bmatrix}$$

$$\begin{bmatrix} \Delta_1 \\ \Delta_2 \\ \Delta_3 \end{bmatrix}=\frac{F_P l}{EA}\begin{bmatrix} -1 \\ -1 \\ -2-2\sqrt{2} \end{bmatrix},$$

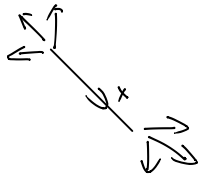
$$\overline{F}^{①}=F^{②}=k^{②}\delta^{②}=\frac{EA}{l}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ -1 \\ -2-2\sqrt{2} \end{bmatrix}\frac{F_P l}{EA}=\begin{bmatrix} F_P \\ 0 \\ -F_P \\ 0 \end{bmatrix}$$

$$F^{①}=k^{①}\delta^{①}=\frac{EA}{l}\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \end{bmatrix}\frac{F_P l}{EA}=\begin{bmatrix} 0 \\ -F_P \\ 0 \\ F_P \end{bmatrix},$$

$$\overline{F}^{①}=TF^{①}=\begin{bmatrix} F_P \\ 0 \\ -F_P \\ 0 \end{bmatrix}$$

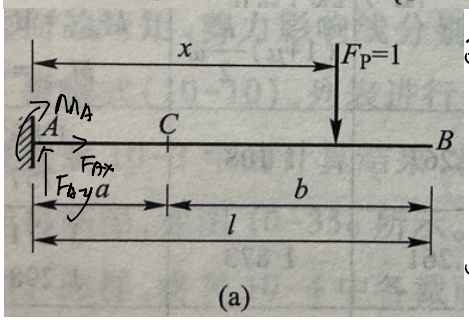
$$F^{③}=k^{③}\delta^{③}=\frac{EA}{\sqrt{2}l}\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}\begin{bmatrix} 0 \\ -1 \\ -1 \\ -2-2\sqrt{2} \end{bmatrix}\frac{F_P l}{EA}=\begin{bmatrix} -F_P \\ F_P \\ F_P \\ -F_P \end{bmatrix}$$

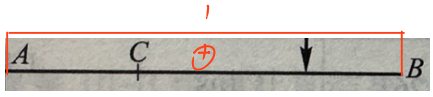
$$\overline{F}^{③}=TF^{③}=\begin{bmatrix} -\sqrt{2}F_P \\ 0 \\ \sqrt{2}F_P \\ 0 \end{bmatrix}$$



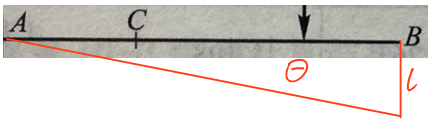
10-1 用静力法作影响线：

(a) F_{Ay}、M_A、M_C 及 F_{QC} 的影响线；

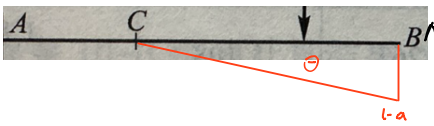


① 

F_{Ay} · I.L

② 

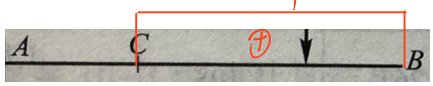
M_A · I.L

③ 

M_C · I.L

$\sum M_A=0$: $-M_A-F_P\cdot x=0$ $M_A=-x$

$\sum M_C=0$: $-M_C-F_P(x-a)=0$ $M_C=-x+a$

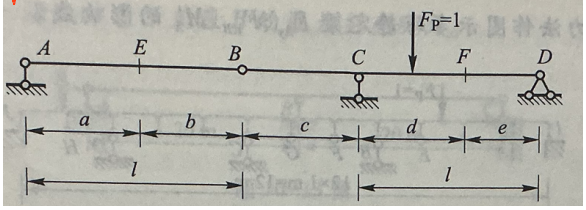
④ 

F_{QC} · I.L

2025年11月19日

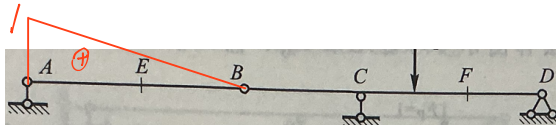
10-2 用静力法作下列多跨静定梁 F_{RA}、M_E、F_{QE}、F_{RC}、M_F 及 F_{QF} 影响线。

10-2



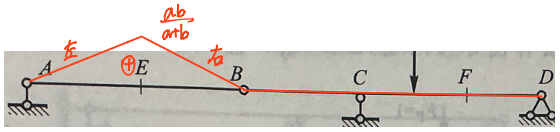
多跨静定梁得先判断基本(静定自身)、附属结构(依靠静定)，附属结构所受荷载会传递到基本结构，但基本传不到附属。

解：

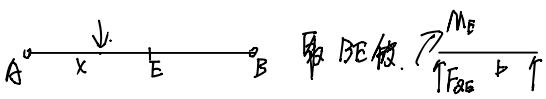
① 

F_{RA} · I.L

F_{RA}：AB为附属结构，BD为基本结构，则当 F_P 作用于BD时，$F_{RA}=0$；于AB段时，F_P 作用于A，则 $F_{RA}=1$，F_P 作用于B则 $F_{RA}=0$

② 

M_E · I.L

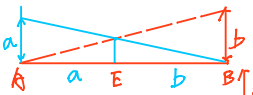


取BE段，$-X+F_{RB}(a+b)=0$，$F_{RB}=\frac{x}{a+b}$

则 $F_{RBmax}=\frac{a}{a+b}$

$F_{RBmax}\cdot b-M_E=0$ $M_E左=\frac{Xb}{a+b}$

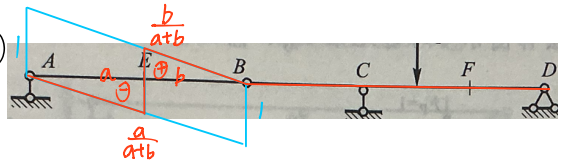
若用画图方法，则为



①当F于AE上运动时，当F于A时，$F_{RB}=0$

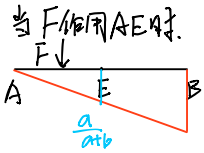
则 $M_{E\max}=b$，此时F于B作图只保留AE部分。

②F于EB上运动时，当F于B时 $F_{RB}=F=1$

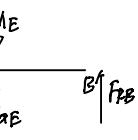
③ 

F_{AE} I.L

当F作用AE时，



取BE段

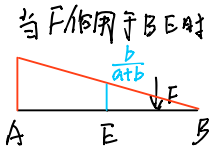


$F_{AE}=-F_{RB}$

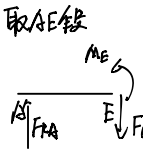
当F作用于B时，$F_{RB}=F=1$

当F作用于A时，$F_{RB}=0$

当F作用于BE时



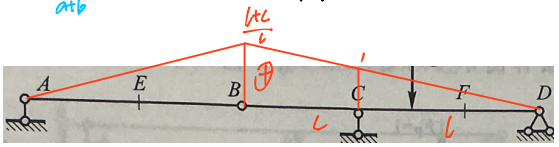
取AE段



$F_{RE}=F_{RA}$

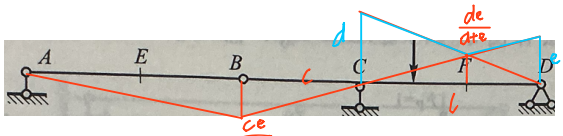
当F作用于A时，$F_{RA}=F=1$

当F作用于B时，$F_{RA}=0$

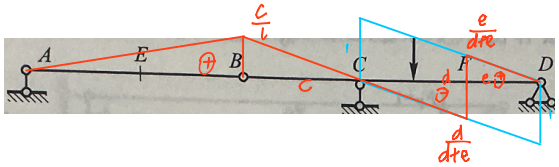
④ 

F_{RC} · I.L

$X=\frac{l+c}{l}$

⑤ 

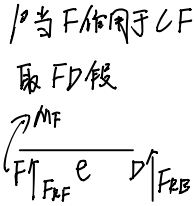
M_F · I.L

⑥ 

F_{QF} · I.L

①当F作用于CF时

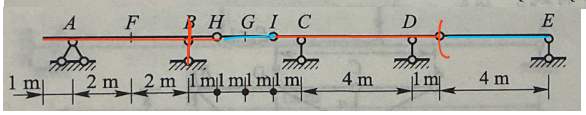
取FD段



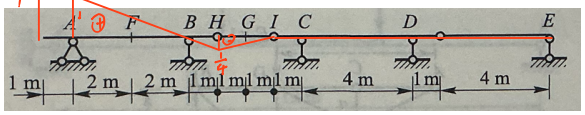
(①②③)中外伸梁部分直接延长简支梁上的即可。除非是求支座反力，否则支座处影响线为0

影响线由基石附部分作到附属部分。求附属部分上点的影响线，不考虑基础部分。

10-3 用静力法作下列多跨静定梁 F_{RA}、F_{QB}^{L}、F_{QB}^{R}、M_F、F_{QF}、M_G 及 F_{QG} 影响线。



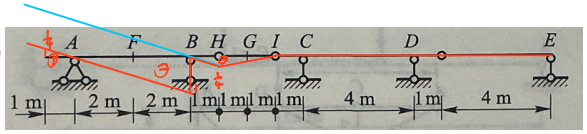
(AH为基础部分，HI与DE为附属部分。CD段是因为该结构受竖直方向荷载，水平方向不需要约束，CD正好也有竖直方向约束，则CD部分为基础部分)

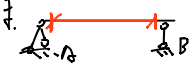
①  F_{RA} 影响线

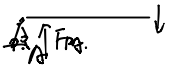
在HI段。F作用于H时，F全传到基础部分AB，则HI段H处最大。作用于I时，F全传到基础部分CD，此时让 $F_{RA}=0$

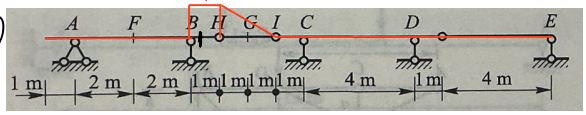
在IE段时，F不会影响到A，则全为0。

相当于，基础A及与其相连的附属为一体，与基础B及其附属独立

②  $F_{QB}^{左}$ 影响线

当F作用于A右、B左侧时  相当于红色部分 取"$\sum M_B$" 在F为负，当F于A有 $F_{RB}=0$，F于B有 $F_{RB}=1$

当F作用于B右侧时 取  F于A时 $F_{RA}=1$，F于B时 $F_{RA}=0$

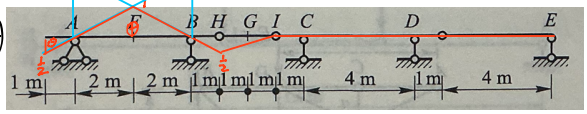
③  $F_{QB}^{左}$ 影响线

B左时，参考简支梁。B右时参考悬臂梁

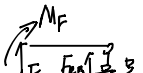
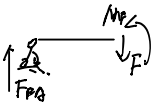
悬臂梁固定端左侧无剪力

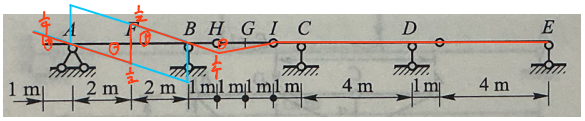
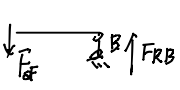
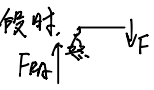
当F于"B右，H"段时，隔离体为 则 $F_{QB}^{左}=F=1$，简支梁的伸梁相当于悬臂梁，即

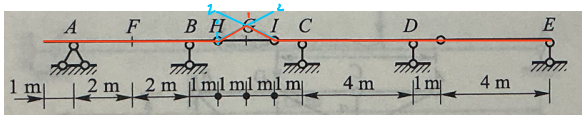
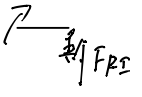
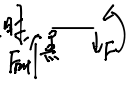
"为右"的左侧截面无剪力。

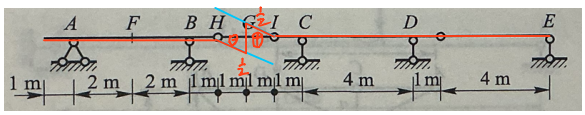
④  M_F 影响线 (下)

方向为下侧受拉为正，为正时影响线画在上方。

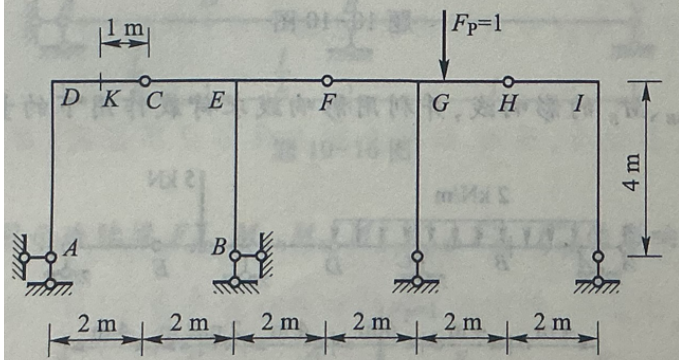
1° F于AF段时  $M_F=F_{RB}\cdot 2m$ 2° F于BF段时 

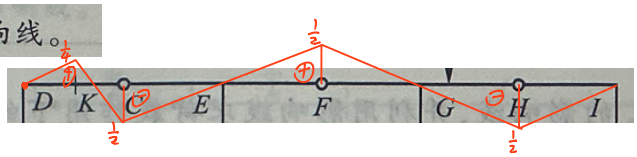
⑤  F_{QF} 影响线 1° F于AF段时  2° F于BF段时 

⑥  M_G 影响线 1° F于HG时  2° F于GI时 

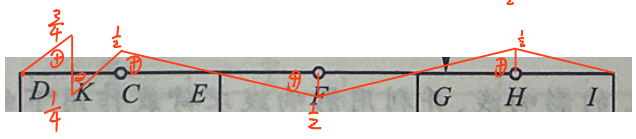
⑦  F_{QG} 影响线 1° F于HG时

10-5 用静力法作刚架中 M_K、F_{QK}及 F_{NK}的影响线。

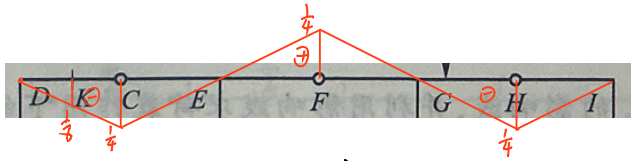




M_K.I.L



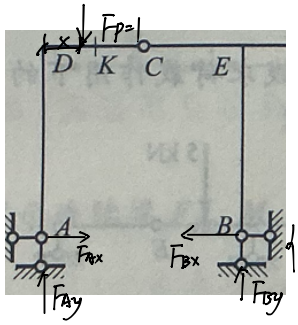
F_{QK}.I.L



F_{NK}.I.L

解：基本结构为DF，附属结构为FH与HI。

① F_P作用于DC时：



$\sum M_B=0$: $F_P(4-x)-F_{Ay}\cdot 4=0$

$\sum F_y=0$: $F_{Ay}+F_{By}-F_P=0$

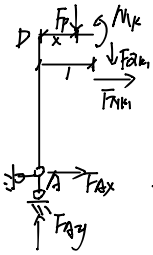
$F_{Ay}=\frac{4-x}{4}$ $F_{By}=\frac{x}{4}$, $x\in[0,12m]$

$\sum M_C=0$: $F_{By}\cdot 2-4F_{Bx}=0$

$F_{Bx}=\frac{x}{8}(\leftarrow)$ $F_{Ax}=\frac{x}{8}(\rightarrow)$

$x\in[0,2]$m

当F_P作用于K左时：



$F_{Ay}=\frac{4-1}{4}=\frac{3}{4}$, $F_{Ax}=\frac{1}{8}$

$\sum M_K=0$: $-\frac{1}{8}\times 4+\frac{3}{4}\times 1=\frac{1}{4}=M_K$

$\sum F_y=0$: $F_{Ay}-F_{QK1}=0$, $F_{QK1}=\frac{3}{4}$

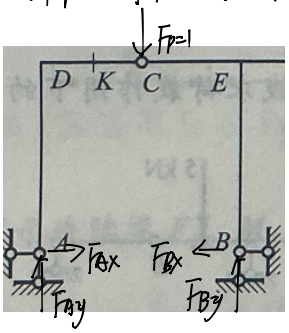
$\sum F_x=0$: $F_{Ax}+F_{NK1}=0$, $F_{NK1}=-\frac{1}{8}$

若F_P作用于K右侧时：

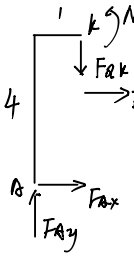
$M_K=\frac{1}{4}$, $F_{NK}=\frac{1}{8}$, $\sum F_y=0$: $F_{Ay}-F_P-F_{QK2}=0$

$F_{QK2}=\frac{3}{4}$

③当F_P作用于C时，即F_P作用于C时



$F_{Ax}=F_{Bx}=\frac{1}{4}$, $F_{Ay}=F_{By}=\frac{1}{2}$



$\sum M_K=0$: $-F_{Ay}\cdot 1+F_{Ax}\cdot 4+M_K=0$

$M_K=-\frac{1}{2}$

$\sum F_y=0$: $F_{Ay}-F_{QK}=0$, $F_{QK}=\frac{1}{2}$

$\sum F_x=0$: $F_{NK}+F_{Ax}=0$, $F_{NK}=-\frac{1}{4}$

③当F_P作用于CF即E时：

由于B为支座，则

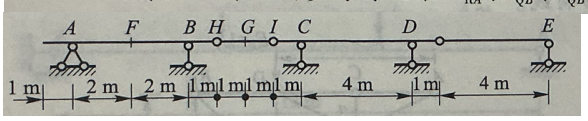
$F_{QK}=F_{NK}=M_K=0$

由CF，FH，HI并未内部断开，则在这3段内为直线

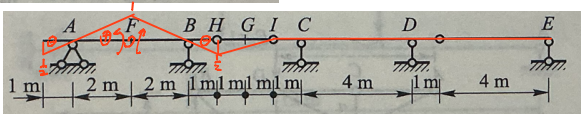
G，I为支座同理。

10-6 用机动法做习题10-3。

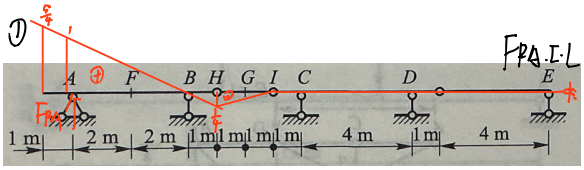
10-3 用静力法作下列多跨静定梁 F_{RA}、F_{QB}^{L}、F_{QB}^{R}、M_F、F_{QF}、M_G及F_{QG}影响线。



④ M_F.I.L

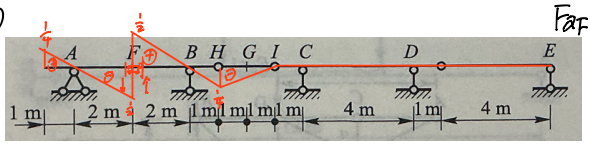


解：①



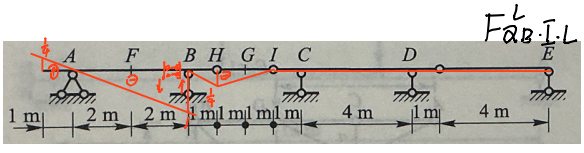
F_{RA}.I.L

⑤



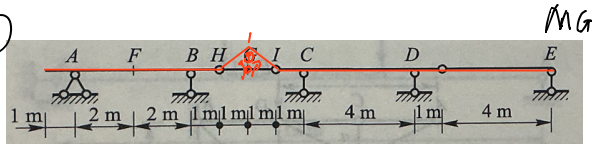
F_{QF}.I.L

②



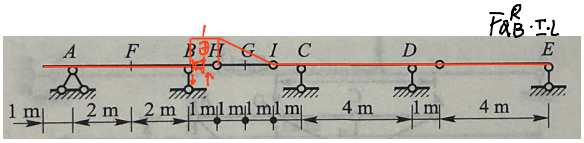
F_{QB}^{L}.I.L

⑥



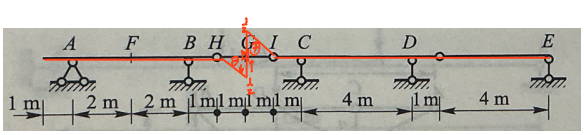
M_G

③



F_{QB}^{R}.I.L

⑦



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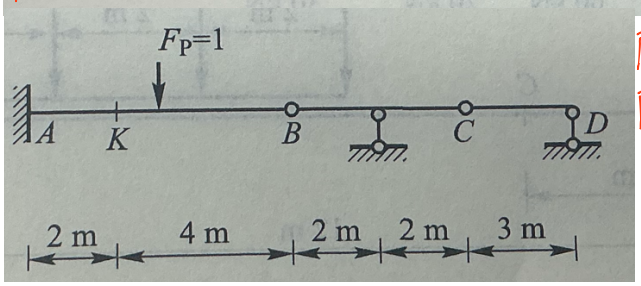
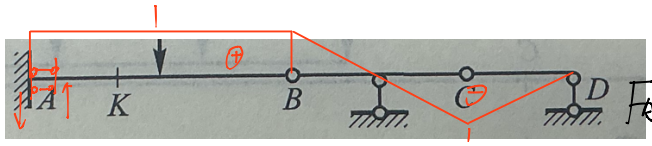
10-7 用机动法作多跨静定梁中 F_{Ay}、M_A、M_K、F_{QK} 的影响线。

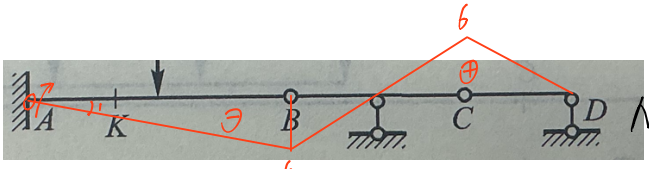


附属部分随基本部分的移动而移动，不会约束基本部分。
附属部分支座点影响线坐标为0

解：



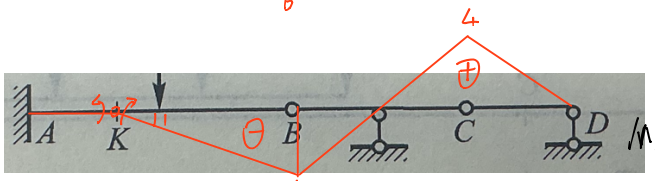
F_{Ay} 影响线



M_B 影响线

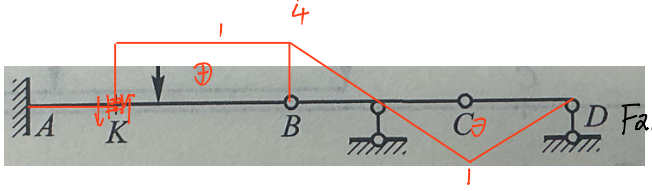
A只能转动因为“链”，则只能B点下降。

$6=\tan\theta\cdot$ 内引，$\tan\theta=1$



M_K 影响线

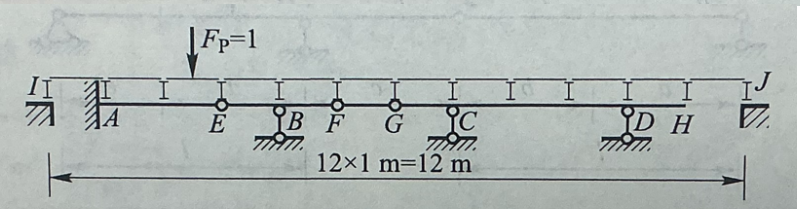
AK段几何不变，不会发生转角



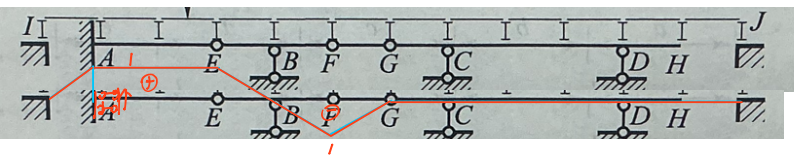
F_{QK} 影响线

“上负下正”让包角杆顺时针转动为正方向

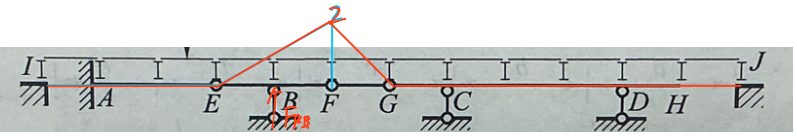
10-8 用静力法作图示多跨静定梁 F_{Ay}、F_{RB}、M_A 的影响线。



解：AE与GH为基本部分，EF、FG为附属部分



F_{Ay} 影响线



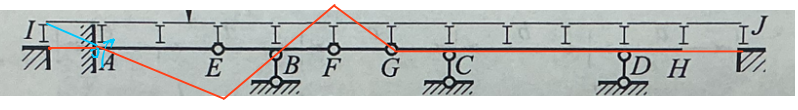
F_{RB} 影响线

EF 附属于AE，FG附属于GH和AF

则能影响B处只有EF与FG。

（只能被上层影响、同层影响）

F G
EF GH
A E



M_A 影响线

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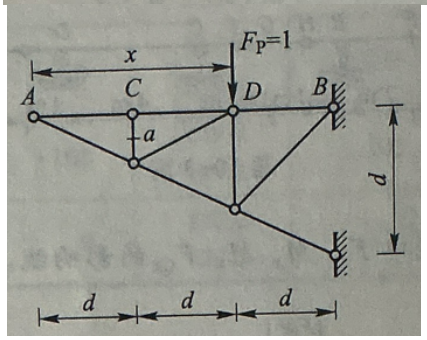
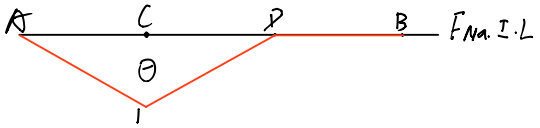
10-9 10-9 作图示桁架 F_{Na} 的影响线。

桁架的影响线必然为间接荷载，或者叫结构荷载，因为不会于杆件上受载只会在结点上受载。

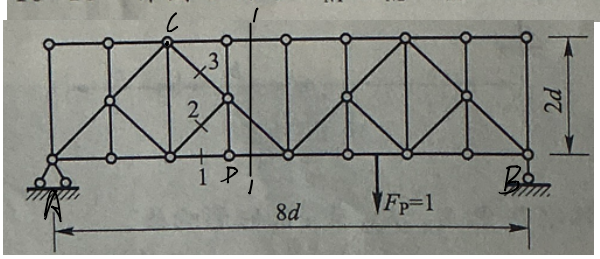


则当 F_P 作用于 C 时，$F_{NA}\neq 0$，当 F_P 作用于 A、D、B 时 $F_{NA}=0$ 可此步为直接荷载定点。然后再在点与点之间连直线。

桁架拉正压负。则 F_{NA} 的影响线为



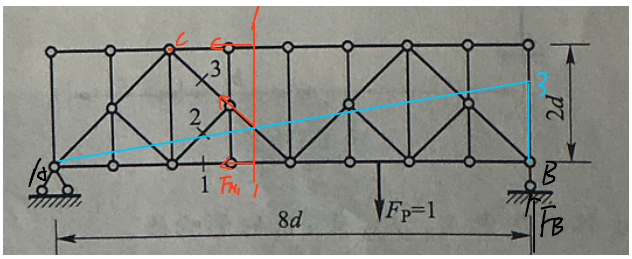
10-10 作图示桁架 F_{N1}、F_{N2}、F_{N3} 的影响线。



解：①求 F_{N1} 的影响线：

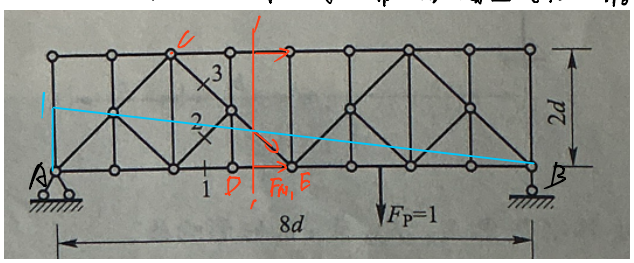
D结点处两横杆受的力等大等号，故将结构沿 I-I 截面分开，对C点求矩。

1°当 F_P 作用于 I-I 截面左侧时，取 I-I 截面右侧分离体。



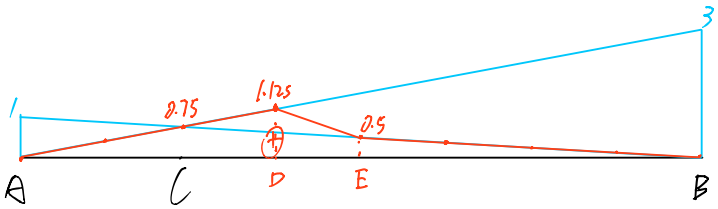
$\sum M_C=0$：$-F_{N1}\cdot 2d+F_{By}\cdot 6d=0$，$F_{N1}=3F_{By}$

2°当 F_P 作用于 I-I 截面右侧时，取 I-I 截面左侧分离体。

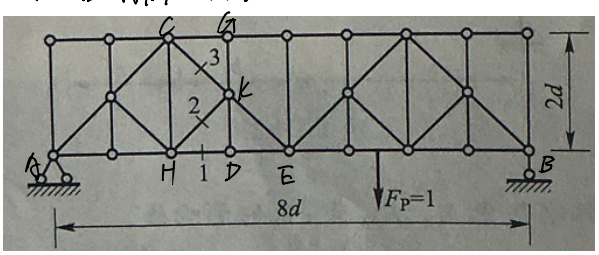


$\sum M_C=0$：$F_{N1}\cdot 2d-F_{Ay}\cdot 2d=0$，$F_{N1}=F_{Ay}$

则 F_{N1} 的影响线为



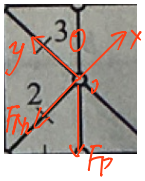
②求 F_{N2} 的影响线.



1°当 F_P 作用于 AH 段与 EB 段时，对 K 结点进行受力分析

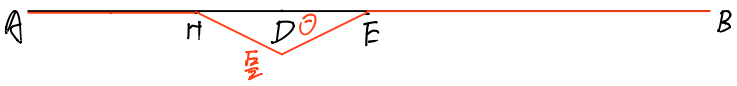
则 对 G、D 结点 GK 杆与 DK 杆为零杆。此时，K 结点为 T 形结点，$F_{N2}=0$

2°当 F_P 作用于 D 时，对 K 结点进行受力分析

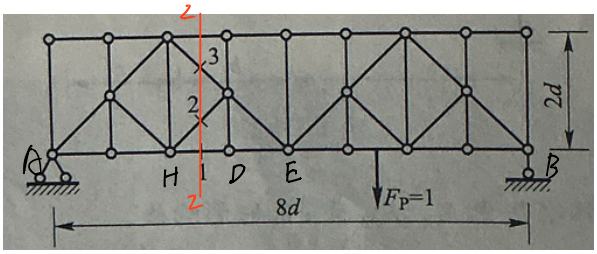


则 $F_{N2}+\frac{1}{2}F_P=0$，$F_{N2}=-\frac{1}{2}$

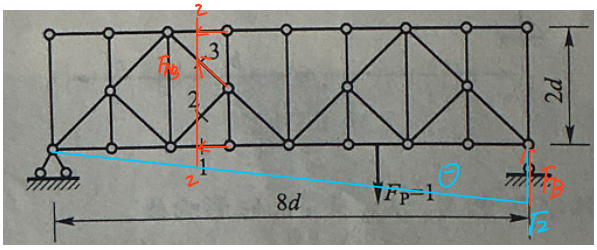
则 F_{N2} 的影响线为



③ 求 F_{N3} 的影响线.

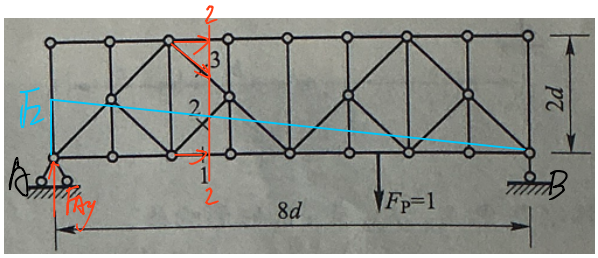


1° 当 F_P 在 AH 段上时，取 2-2 截面右侧分离体



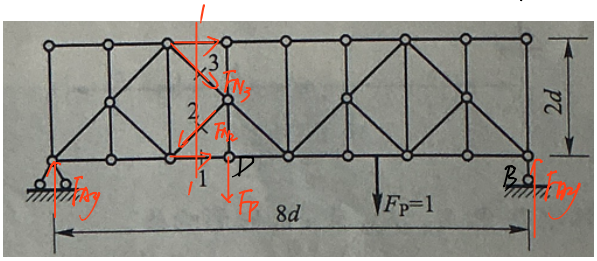
$\sum F_y=0$: $F_{By}+F_{N3}\sin 45°=0$, $F_{N3}=-\sqrt{2}F_{By}$

2° 当 F_P 作用于 EB 段时，取 2-2 截面左侧隔离体.



$\sum F_y=0$: $F_{Ay}-\sin 45°F_{N3}=0$, $F_{N3}=\sqrt{2}F_{Ay}$

3° 当 F_P 作用于 D 时，取 2-2 截面左侧分离体.

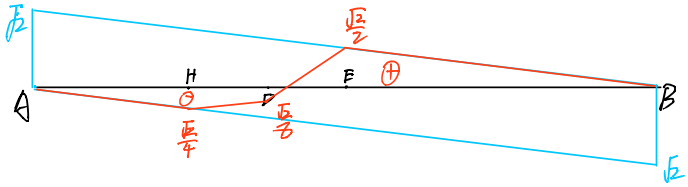


对整体：$\sum F_y=0$: $F_{Ay}+F_{By}=F_P$, $\sum M_D=0$: $-F_{Ay}\cdot 3d+F_{By}\cdot 5d=0$

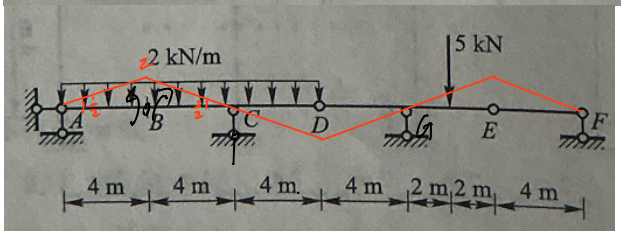
则 $F_{Ay}=\frac{5}{8}F_P=\frac{5}{8}$

对分离体：$\sum F_y=0$: $F_{Ay}-\frac{\sqrt{2}}{2}F_{N2}-\frac{\sqrt{2}}{2}F_{N3}=0$, $F_{N3}=\frac{5}{8}$

则 F_{N3} 的影响线为：



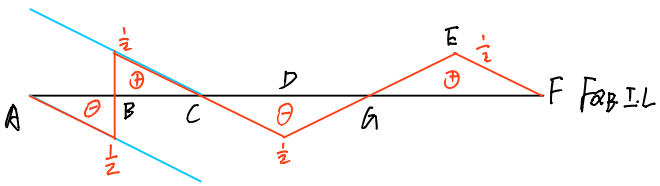
10-11 作 F_{QB}、M_B 的影响线，并利用影响线求荷载作用下的量值。

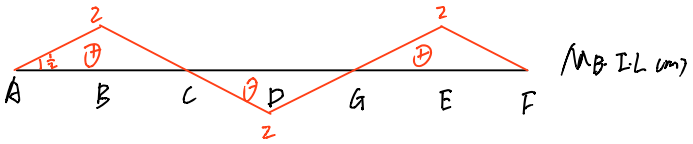


F_{QB} 为均布荷载乘以影响线图形面积 + 竖标 × 集中力

M_B = 均布荷载 × 影响线图形面积（有正负可相消）+ 竖标 × 集中力

解：AD 为基本结构，DE 与 EF 为附属结构.





则 $F_{QB}=\frac{1}{2}\times 4\times\frac{1}{2}\times 2+\frac{1}{4}\times 5=-0.75\,kN$　　$M_B=\frac{1}{2}\times 2\times 4\times 2+5\times 1=13\,kN\cdot m$

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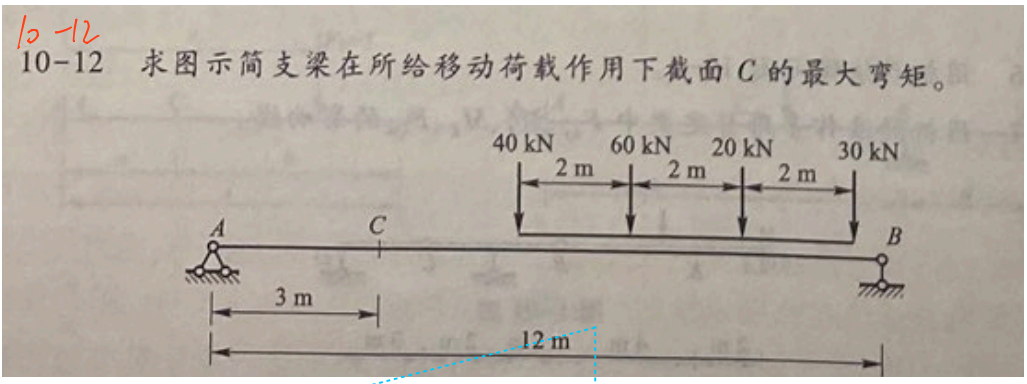
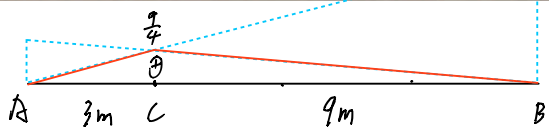
10-12

10-12 求图示简支梁在所给移动荷载作用下截面 C 的最大弯矩。



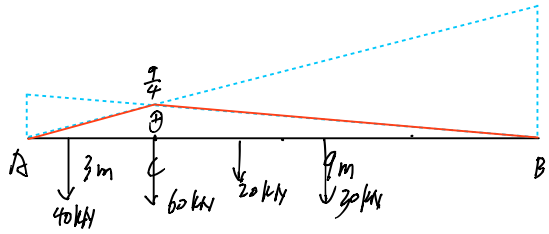
①先做 M_C I.L

解:



M_C I.L

① 将 60 kN 置于C点

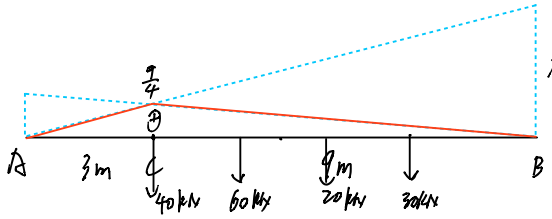


左移: $\frac{40+60}{3}=\frac{100}{3}>\frac{20+30}{9}=\frac{50}{9}$

右移: $\frac{40}{3}>\frac{60+20+30}{9}=\frac{110}{9}$

舍去.

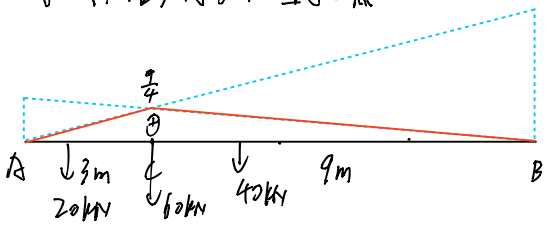
② 将 40 kN 置于C点.



左移: $\frac{30}{3}>\frac{60+20+30}{9}=\frac{110}{9}$

右移: $0<\frac{40+60+20+30}{9}$, 则 $M_{C1}=\frac{9}{2}\times40+\frac{7}{2}\times60+\frac{5}{2}\times20+\frac{3}{2}\times30=242.5\,kN\cdot m$

③ 车辆调头后, 将 60 kN 置于C点



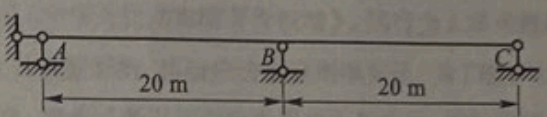
左移: $\frac{20+60}{3}>\frac{40}{9}$

右移: $\frac{20}{3}<\frac{60+40}{9}$, 则 $M_{C2}=\frac{3}{2}\times20+60\times\frac{9}{2}+\frac{7}{2}\times40=220\,kN\cdot m$

$M_{C\max}=\max(M_{C1},M_{C2})=242.5\,kN\cdot m$

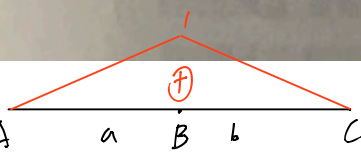
10-13

10-13 求中—活载作用下支座 B 的最大支座反力。



题 10-13 图

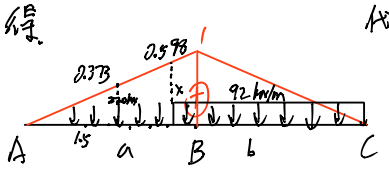
解:



F_{RB} I.L

由 $\frac{F_R^L}{a}=\frac{F_R^R}{b}$ 得

$\frac{F_RL}{20}=\frac{F_R^R}{20}$ ①



$F_R^L=220\times5+92x=(1100+92x)$ kN

$F_R^R=92\times20=1840$ kN

代入方程①得: $1100+92x=1840$

$x=8.04$ m

正中间的 220 kN 对应纵标为

$y=\frac{20-(3\times1.5+8.04)}{20}=0.373$

均布荷载边缘对应的纵标为 y_2:

$y_2=\frac{20-8.04}{20}=0.598$

$F_{RB\max}=5\times220\times0.373+\frac{(0.598+1)\times8.04}{2}\times92+\frac{92\times11.20}{2}=1921.30$ kN

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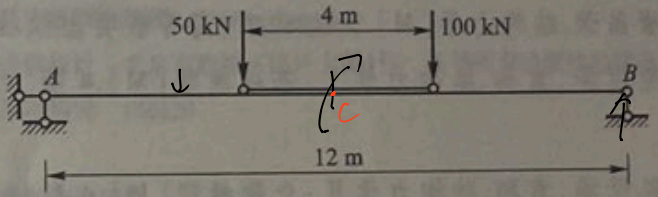
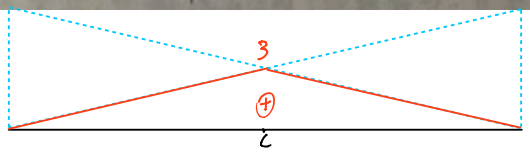
10-14 求图示简支梁的绝对最大弯矩，并与跨中截面的最大弯矩相比较。



简支梁最大弯矩 要将最大力与合力关于中点对称布置

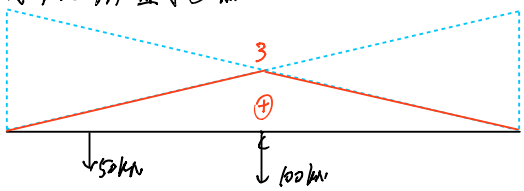
绝对最大从作用于100kN作用位置

解：



$M_{C.I.L}$

① 将100kN置于C点



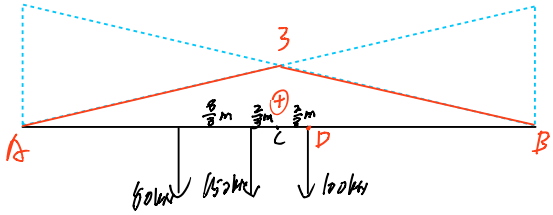
左极：$\frac{50+100}{6}>0$ 右极：$\frac{50}{6}<\frac{100}{6}$ 成立。

$M_{C.I.L}$ $M_{Cmax}=100\times3+50\times1=350\ kN\cdot m$ 即跨中截面最大弯矩为350kN·m

② 简支梁的合力及其作用位置 (合力对1点的矩为各分力对该点力矩之和)

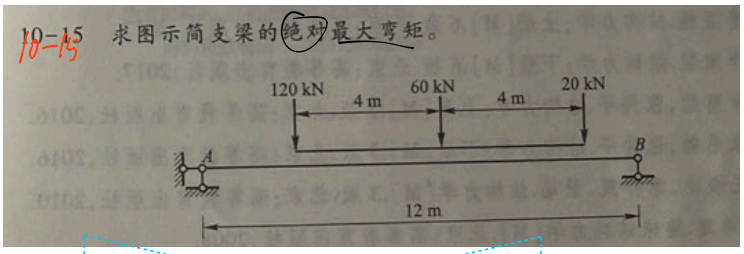
对100kN作用点取矩：$150a=50\times4+100\times0$ 则 $a=\frac{4}{3}\ m$

$M_{max}=\frac{150\times(\frac{2}{3}+6)^2}{12}-50\times4=355.56\ kN$



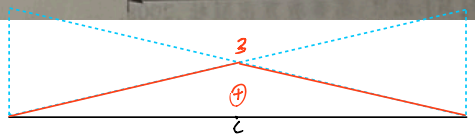
绝对最大从作用于100kN作用位置，即D点。先算合力时支座B的弯矩，除总长得F_{RA}，让F_{RA}与50kN对D点求矩，这矩为M_{max}。

10-15 求图示简支梁的绝对最大弯矩。



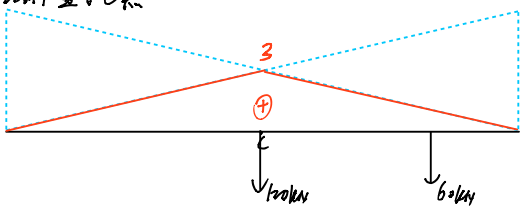
必须先把力放在中点 判断是否为极值 后续才能据此算合力。

解：



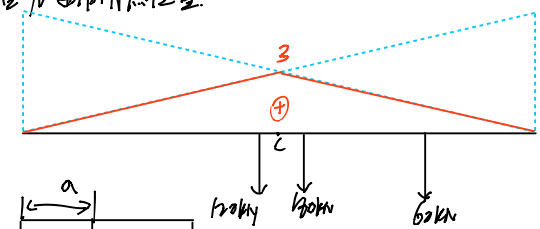
$M_{C.I.L}$

① 将120kN置于C点

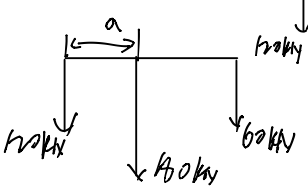


左极：$\frac{120}{6}>\frac{60}{6}$ 右极：$0<\frac{60\times2}{6}$ 成立

② 求合力与作用点位置



$M_{max}=\frac{180\times(6+\frac{2}{3})^2}{12}-60\times4=426.67\ kN\cdot m$



对120kN作用点求矩：

$180a=60\times4$ $a=\frac{4}{3}\ m$